



Firm-Level Credit Risk Early-Warning System

Human Rules vs Statistical Models vs Machine Learning



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Tools: Python, pandas, scikit-learn, Excel, S&P Capital IQ, FRED

Executive Summary

The objective of this project is to develop a forward-looking firm-level credit risk early-warning system and to evaluate how different approaches to credit assessment compare in practice. Specifically, the project contrasts traditional human-style rule-based credit scoring, classical statistical models, and machine-learning methods in terms of predictive performance, robustness, and interpretability.

Rather than focusing solely on maximizing predictive accuracy, the analysis emphasizes realistic modeling assumptions, proper time-based validation, and economically interpretable results. This reflects how credit risk models are evaluated and deployed in real-world settings such as corporate lending, portfolio risk management, and financial planning and analysis.

Data and Scope

The model is built using firm-level financial data sourced from S&P Capital IQ, combined with lagged macroeconomic indicators from public sources. The dataset is structured as a firm-year panel and designed to predict one-year-ahead default risk using only information available at the time of prediction. Default is proxied using a forward-looking financial distress indicator, consistent with early-warning system design where the goal is to identify deterioration prior to formal default events.

Modeling Approaches

Three modeling approaches are implemented and evaluated under a consistent framework:

- A rule-based Altman-style score representing human heuristic credit assessment
- A logistic regression model representing classical statistical credit modeling
- A Random Forest model representing machine-learning-based credit risk modeling

All models use the same underlying data, features, and validation horizon to ensure fair comparison.

Key Results

Model	Description	AUC	KS
Altman-Style Rules	Human / heuristic benchmark	~0.55	~0.14
Logistic Regression	Classical statistical model	~0.64	~0.28
Random Forest	Machine learning model	~0.71	~0.38

Core Insight

The results show that most gains in firm-level default prediction arise from moving away from heuristic rules toward statistically estimated models. Logistic regression captures the majority of available predictive signal by optimally weighting financial ratios and macroeconomic variables. Machine learning further improves performance by modeling nonlinear relationships and missing-data patterns, though the gains are incremental rather than transformative.

Business Relevance

The resulting early warning system supports practical credit risk applications, including portfolio monitoring, risk bucketing, credit review prioritization, and identification of high-risk firms for further analysis. The framework closely mirrors how credit risk, FP&A, and strategy teams translate financial and macroeconomic data into actionable risk insights.

Business Problem Definition

Accurate and timely identification of deteriorating credit quality is a core requirement for banks, lenders, and credit risk teams. Firm-level early-warning systems are used to flag companies that may default in the near future, allowing institutions to adjust exposure, initiate credit reviews, or apply risk mitigation strategies. Historically, credit assessment relied heavily on expert judgment and simple financial ratio rules. While intuitive, these approaches often fail to consistently rank firms by risk, especially across economic cycles.

Modern credit risk management increasingly relies on data-driven models, but there remains skepticism around machine learning due to concerns about interpretability, stability, and overfitting. As a result, a key practical question is not whether machine learning “works,” but how much additional value it provides relative to simpler, more transparent models.

This project is explicitly designed to answer that question. By structuring the analysis as a comparison between human-style rules, classical statistics, and machine learning using the same data, validation horizon, and evaluation metrics the project provides an honest and realistic assessment of the trade-offs involved in deploying more complex credit models.

Data Overview & Sources

The analysis combines firm-level financial data with macroeconomic indicators to construct a forward-looking credit risk dataset.

Firm financial data are sourced from S&P Capital IQ, using annual reported fundamentals for publicly listed firms. These include balance sheet, income statement, and cash flow items that are commonly used in corporate credit analysis. Each firm is uniquely identified using a stable entity identifier to ensure consistent tracking across years.

Default events are proxied using a forward-looking distress indicator, defined at the firm-year level. A firm is considered to default at time $t + 1$ if it exhibits sustained financial distress following the reporting period at time t . This approach aligns with early-warning system design, where the goal is to anticipate deterioration rather than react to legal bankruptcy filings.

Macroeconomic data are incorporated to capture systematic risk effects. These variables include GDP growth, the unemployment rate, and a financial conditions index, all lagged to ensure that only information available at the time of prediction is used.

The final dataset contains approximately 10,000 firm-year observations, spanning multiple economic cycles, and is structured as a balanced panel suitable for time-based model validation.

Data Cleaning & Preprocessing

Significant preprocessing was required to transform raw vendor exports into a clean modeling dataset. Financial data exported from S&P Capital IQ often includes metadata rows, multi-level headers, duplicated fields, and sparse reporting across firms. These issues were addressed through a systematic cleaning pipeline.

First, non-data rows and metadata were removed, and column headers were standardized to ensure consistent naming conventions. Firm identifiers were validated to ensure a one-to-one mapping between firms and observations. When merging multiple financial tables, duplicate columns arising from overlapping exports were explicitly resolved.

The dataset was reshaped from a wide, multi-year format into a firm-year panel structure, enabling proper time-series alignment and forward-looking labeling. Observations with insufficient historical information were excluded to prevent biased estimates.

Outliers in financial ratios were addressed using winsorization at the 1st and 99th percentiles. This step reduces the influence of extreme values without discarding economically meaningful observations, which is standard practice in credit modeling.

Missing data were handled explicitly rather than ignored. Instead of dropping observations, missing-value indicator variables were created for key financial ratios. This approach allows models to learn from the fact that missing disclosures themselves may signal elevated credit risk, a well-documented phenomenon in corporate credit analysis.

Feature Engineering

Feature engineering focused on constructing economically meaningful and interpretable predictors that reflect a firm's financial health and exposure to macroeconomic conditions.

Core financial ratios include measures of profitability (return on assets), leverage, debt service capacity (interest coverage), and liquidity (operating cash flow relative to debt). These ratios are widely used in both academic research and industry credit models.

Macroeconomic variables were lagged by one period to ensure strict temporal ordering. This prevents look-ahead bias and reflects the reality that credit decisions are made based on known economic conditions, not future realizations.

Missing-value indicators were added for each major financial ratio. These binary flags allow models, particularly machine learning models to distinguish between firms with genuinely strong metrics and firms whose data are incomplete or unavailable.

All variables were standardized where appropriate, particularly for logistic regression, to ensure numerical stability and comparability across features.

Modeling Approach

Three distinct modeling approaches were implemented to represent different philosophies of credit risk assessment.

The Altman-style rule-based score serves as a human benchmark. It combines financial ratios using fixed, hand-selected weights based on economic intuition. No parameters are learned from the data, making this approach fully transparent but limited in flexibility.

The logistic regression model represents the classical statistical approach to credit modeling. It estimates the probability of default as a function of the engineered features, learning optimal linear coefficients from the data while maintaining interpretability and stability.

The Random Forest model represents a machine-learning approach. By combining multiple decision trees, it captures nonlinear relationships, interaction effects, and complex missing-data patterns. Conservative hyperparameters were chosen to reduce overfitting and align with credit risk best practices.

All models were trained using a time-based train/test split to simulate real-world forecasting conditions.

Model Evaluation & Validation

Model performance was evaluated using out-of-sample AUC and KS statistics. These metrics assess a model's ability to rank firms by default risk rather than predict exact probabilities.

The Altman-style score achieved an AUC of approximately **0.55**, indicating limited discrimination slightly above random guessing. Logistic regression improved performance substantially, achieving an AUC of approximately **0.64** and a KS statistic of **0.28**, consistent with industry benchmarks for firm-level PD models.

The Random Forest model delivered the strongest performance, with an AUC of approximately **0.71** and a KS statistic of **0.38**. ROC curves and risk-bucket analysis confirm that the machine-learning model provides superior separation between defaulting and non-defaulting firms, particularly in the highest-risk deciles.

Importantly, performance gains from machine learning are incremental rather than extreme, reinforcing the importance of strong feature engineering and economic intuition.

Risk Buckets and Practical Interpretation

To translate model outputs into actionable insights, predicted probabilities of default from the Random Forest model were grouped into decile-based risk buckets. Firms in Bucket 0 represent the lowest risk segment, with very low predicted default probabilities and minimally observed default incidence. As bucket levels increase, both predicted risk and realized default rates rise steadily, with Buckets 7–9 capturing a disproportionate share of default events.

Firms in the highest-risk bucket (Bucket 9) exhibit default rates that are several times higher than those observed in the lowest-risk buckets, indicating strong separation between safe and distressed firms. The monotonic increase in observed default rates from Bucket 0 through Bucket 9 demonstrates that the model produces stable and intuitive risk rankings rather than noisy or unstable classifications.

These risk buckets enable practical credit risk applications, including the construction of monitoring watchlists for high-risk firms, prioritization of credit reviews, and assessment of portfolio concentration in elevated-risk segments. Firm-level outputs were exported for each bucket, allowing direct inspection of companies in the highest risk deciles an essential step in real-world credit risk monitoring and early-warning workflows.

Limitations & Assumptions

Several limitations remain. Default is proxied using a distress indicator rather than legal bankruptcy. Annual data limit short-term forecasting resolution, and results may vary across industries. Future work could incorporate stress-testing scenarios, calibration analysis, and industry-specific models.

Conclusion & Next Steps

This project demonstrates that effective firm-level credit risk modeling is driven primarily by sound economic intuition, disciplined data preparation, and appropriate validation, rather than model complexity alone. A rule-based Altman-style score provides limited discriminatory power, while logistic regression captures most of the available signal by statistically weighting firm fundamentals and macroeconomic conditions. Machine learning, implemented via a Random Forest model, further improves default discrimination by capturing nonlinear interactions and missing-data patterns, though the gains are incremental rather than transformative.

These findings reinforce a key practical insight: in corporate credit risk, model performance improves monotonically with methodological sophistication, but transparency, robustness, and economic interpretability remain essential. The resulting early-warning system is suitable for portfolio monitoring, risk bucketing, and credit review prioritization.

Building on this foundation, a natural next step is to extend the framework from default prediction to macro-financial forecasting. The planned extension is a Macro-Financial Forecasting Engine that compares traditional econometric models (linear regression, ARIMA, VAR) with machine-learning approaches (Random Forest, gradient boosting, and lightweight sequence models) in forecasting revenues, margins, or cash flows. The emphasis will be on evaluating forecast accuracy, stability across economic regimes, and interpretability under macroeconomic scenarios such as interest-rate hikes or inflation shocks.

Together, these projects form a cohesive analytical toolkit for understanding how macroeconomic conditions propagate through firm fundamentals into financial outcomes and credit risk, closely mirroring real-world applications in FP&A, corporate strategy, credit risk management, and asset management.

Appendix

